

Further Observations on Powerful Numbers: Gaps, Cyclotomic Values, and Independence Tests

Berkay Yüksel Sayim
Independent Researcher, Germany
berksa@tutamail.com · ORCID 0009-0004-4993-7352

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Abstract

We collect observations on powerful numbers that arose in the study of triples and pairs of consecutive powerful numbers. Up to 10^{14} , only 0.10 percent of the gap values between consecutive powerful numbers are powerful, but the most frequent gap values are, and a lemma of Bajpai, Bennett and Chan gives a mechanism for this. We give two explicit infinite chains, each giving powerful values of both $x^2 + x + 1$ and $x^2 - x + 1$, list the powerful values of the cyclotomic polynomials Φ_d with $3 \leq d \leq 12$ up to 10^{12} , and show that $x^4 + 1 = q^3 a^2$ has no solution for primes $q < 2 \cdot 10^{11}$. We test the independence of rare Wieferich-type events along the divisors of an index, with the power of the test, and measure the Wieferich fractions of Pell units against the equidistribution predicted by a conjecture of Katz. Every computed number is generated from the deposited data, and every numbered statement is labeled as proved, taken from the literature, computed, measured, heuristic, or open.

About this series. This paper is one of three parts that share one source, one bibliography, and one numbering. A statement numbered I.4.5 carries that number in its part and in the collected volume. Every number that we computed is generated from the result files of the data deposit or from a cited constant; numbers from the literature carry a citation. The ranges and parameters of a computation and the constants of an argument are written in the text. The open questions of all parts are collected as Problems A to I in the framework introduction of the collected volume and in the file `OPEN_PROBLEMS.md` of the data deposit; both are generated from the parts. This part is a preprint. The author corrects any error that becomes visible after the release in a new version of the record; earlier versions remain available under their own version numbers.

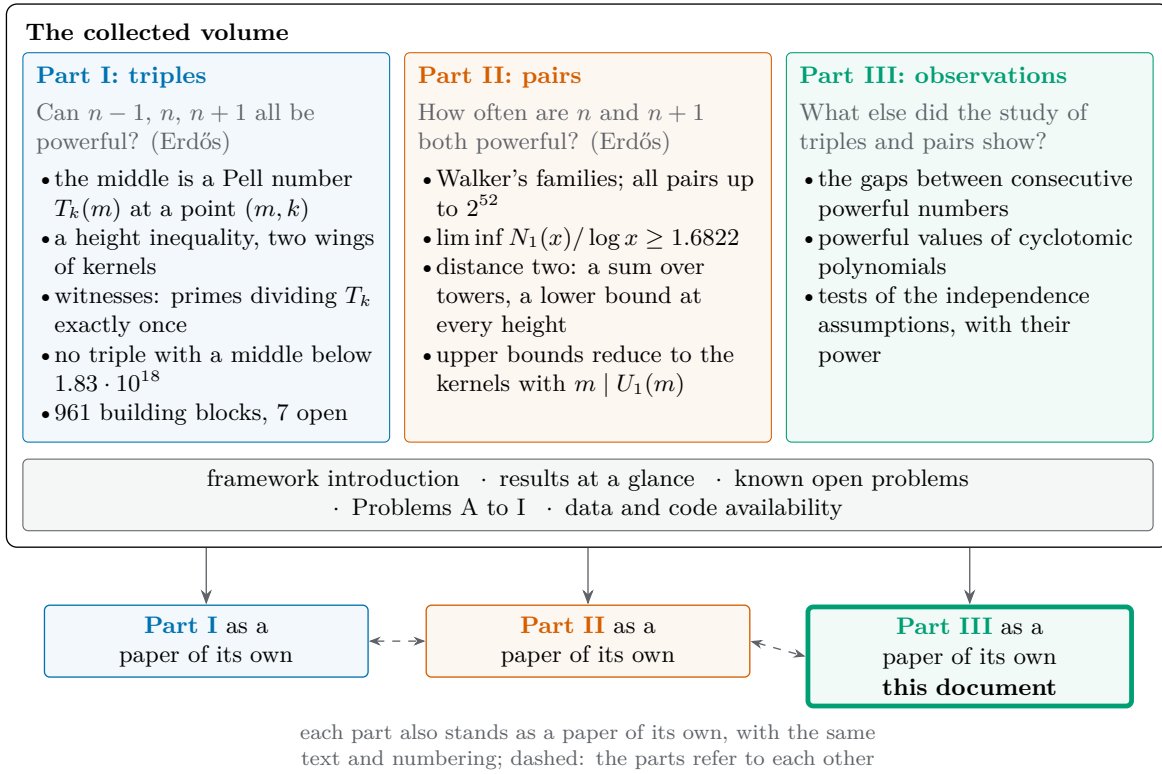


Figure 1: **What this research did.** It followed two questions of Erdős on consecutive powerful numbers, on triples and on pairs, through Pell equations, proved or computed what it could reach, and recorded the rest as observations and open problems. The collected volume contains the three parts and the framework that ties them together; each part also stands as a paper of its own, with the same text and numbering, and the parts refer to each other. The heavier frame marks the document in hand.

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Introduction

Parts I and II study triples and pairs of consecutive powerful numbers. Along the way we measured and proved a number of things that sit at the edge of these two questions and belong to neither. This part collects them. None of them is an attack on the questions of Erdős, and several are consequences or tests of published results; where that is so, the source is named at the statement.

What this part contains. Section 1 looks at the gaps between consecutive powerful numbers: the most frequent gap values up to 10^{14} are powerful, although almost no gap value is, and an elementary lemma gives a mechanism that favors powerful gaps (Proposition III.1.2, Remark III.1.3). Section 2 treats powerful values of cyclotomic polynomials, which arise when a member of a triple is a proper power: two explicit infinite chains, each giving powerful values of both Φ_3 and Φ_6 (Proposition III.2.2), a table up to 10^{12} , and the case $x^4 + 1 = q^3 a^2$ (Proposition III.2.5). Section 3 tests the independence assumption of Part I along the divisors of the index, and states what the test can and cannot see. Section 4 measures the Wieferich fractions of Pell units, the quantity behind the rare events of Part I, against the equidistribution that a conjecture of Katz predicts. Section 5 states two questions.

How to read this part. Every statement is followed by a paragraph *In words*. Every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end); in this part most statements are computations or measurements on a finite range, and they are stated as such. A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Every number we computed is generated from the data files of the deposit.

1 The gap spectrum of powerful numbers

Write the powerful numbers in increasing order and call the difference g of two consecutive members a *gap*; the sequence of gaps is OEIS A076446 [9]. For an integer $g \geq 1$ let $\text{sqfull}(g)$ be the product of the prime powers $p^{v_p(g)}$ with $v_p(g) \geq 2$. Then $\text{sqfull}(g) = g$ holds exactly when g is powerful. Up to 10^{14} there are 21 663 503 powerful numbers and 8 235 300 distinct gap values, and only 0.10 % of these values are powerful; most of them are large and occur once. Among the 62 118 values that occur at least 20 times, 4.81 % are powerful. This section shows that the most frequent gap values are nevertheless powerful, and gives a mechanism that favors them.

Lemma III.1.1 (Bajpai, Bennett and Chan [1, proof of Theorem 1.1]). [literature] *Let n and $n+g$ be powerful. Then $\gcd(n, g)$ is powerful and divides $\text{sqfull}(g)$. In particular $\gcd(n, g) = 1$ if g is squarefree, and $g \mid n$ is possible only if g is powerful.*

Proof. If a prime p divides n and g , it divides $n+g$, so p^2 divides n and $n+g$ and hence their difference g . So every prime of $\gcd(n, g)$ occurs in it at least twice, and with an exponent at most $v_p(g)$, where $v_p(g) \geq 2$. If $g \mid n$, then $\gcd(n, g) = g$ is powerful. $\square$

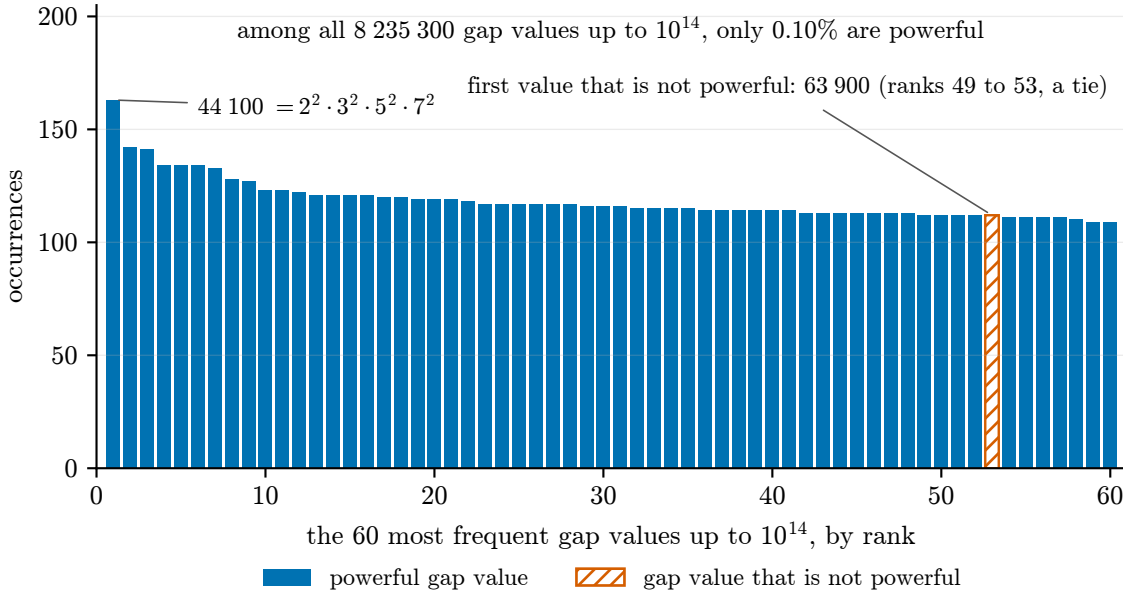


Figure 2: **Up to 10^{14} , the 48 most frequent gap values are all powerful, although only 0.10 % of all gap values are.** The sixty most frequent values of the gap between consecutive powerful numbers up to 10^{14} , by their number of occurrences; values with the same number share their rank and are ordered by size. The first 48 values are all powerful; the first value that is not powerful, 63 900, shares its number of occurrences with powerful values (Proposition III.1.2). Among all 8 235 300 gap values up to 10^{14} , only 0.10 % are powerful. The all-powerful head is shorter at 10^{14} than at 10^{12} (Table 1), so the picture says nothing about larger bounds (Question III.5.1).

That $\gcd(n, g)$ is powerful is used, for k -full numbers and without a separate proof, by Bajpai, Bennett and Chan immediately before their equation (4.2); the consequence for $\text{sqfull}(g)$ is immediate, and it is what we use.

In words. If two powerful numbers are a distance g apart, every prime they share with g appears in g at least twice. So the part of g that such a pair can share is its powerful part; for a gap without square factors there is nothing to share.

Proposition III.1.2 (the head of the gap spectrum). [computed] *Up to 10^{14} the 48 most frequent gap values are all powerful; they are the values that occur more than 112 times. The first value that is not powerful is $63\,900 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 71$, with 112 occurrences, and it is the only one among the 103 values that occur at least as often as the hundredth most frequent. Among the 201 values that occur at least as often as the two hundredth, 191 are powerful. Table 1 gives the same count at 10^{12} and 10^{13} .*

If the 50 most frequent values were drawn at random from all gap values up to 10^{14} , about 0.05 of them would be powerful, and about 2.4 if they were drawn from the values that occur at least 20 times. The most frequent value at 10^{13} and 10^{14} is $44\,100 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$.

bound	powerful numbers	gap values	% of values powerful	most frequent	all powerful to rank	first value not powerful	powerful near rank 200
10^{12}	2 158 391	832 161	0.31	900	69	1656	183 of 215
10^{13}	6 840 384	2 593 517	0.18	44 100	57	63 900	189 of 211
10^{14}	21 663 503	8 235 300	0.10	44 100	48	63 900	191 of 201

Table 1: The gap values between consecutive powerful numbers up to three bounds (Proposition III.1.2). Values of equal frequency share a rank; the column “all powerful to rank” counts the values that occur more often than the first value that is not powerful, and the last column counts the powerful values among those that occur at least as often as the two hundredth most frequent.

In words. Almost no gap value is powerful, but nearly all of the frequent ones are. From 10^{12} to 10^{14} the share of powerful values near the top rises, while the first exception moves up the list; the effect does not simply grow or fade with the bound.

Remark III.1.3 (a ceiling for common divisors). [computed] By Lemma III.1.1, a pair at gap g shares at most $\text{sqfull}(g)$ with g , and only a powerful g can share all of itself. Table 2 lists the 10 values that are not powerful among the 201 values of Proposition III.1.2. Each has a small powerful part that is itself a frequent gap value; values of equal frequency share a rank, so the rank is given as a range. The share of occurrences in which the pair shares the full ceiling is 24.2–44.9 % for the 21 values that occur at least as often as the twentieth most frequent, and 20.8–61.8 % for these 10. So both sides are consistent with one mechanism, a pair sharing the largest divisor that the lemma allows; only the size of that divisor differs.

g	$\text{sqfull}(g)$	rank of $\text{sqfull}(g)$	occurrences of g	% with $\text{gcd} = \text{sqfull}(g)$
63 900	900	9	112	50.9
82 908	1764	36–41	98	49.0
17 100	900	9	97	43.3
23 324	1372	170–175	96	20.8
151 900	4900	10–11	90	45.6
171 900	900	9	90	52.2
7100	100	357–381	89	61.8
26 100	900	9	89	37.1
127 800	1800	29–31	89	53.9
215 100	900	9	89	51.7

Table 2: The frequent gap values up to 10^{14} that are not powerful (Remark III.1.3): all values that are not powerful and occur at least as often as the two hundredth most frequent.

In words. About half of the pairs at gap 63 900 share the factor 900 with the gap, and 63 900 occurs nearly as often as 900 itself. A powerful gap can share all of itself with a pair; this is one reason why the head of the spectrum is powerful, not a proof that it must be.

Remark III.1.4 (a counting model, measured). [measured] Write $g = \text{sqfull}(g) \cdot c$ with c squarefree, and let $\text{occ}(g)$ be the number of occurrences of g up to 10^{14} . Over the 62 118 gap values with at least 20 occurrences, which have 2986 distinct parts $\text{sqfull}(g)$ and 9752 distinct cofactors c , we fitted $\log \text{occ}(g) = u(\text{sqfull}(g)) + v(c)$ with one free value per part and per cofactor. Evaluated on the values held back from the fit, in five random splits of about four

fifths to one fifth over the gap values, with the mean effect used for a part or cofactor that the fit has not seen and an affine recalibration fitted on the training part, the model accounts for 0.545 ± 0.025 of the variance of $\text{occ}(g)$ and 0.514 ± 0.016 of the variance of $\log \text{occ}(g)$, against 0.306 ± 0.005 for a regression of $\text{occ}(g)$ on plain arithmetic features of g on the same values; the margin is positive in each split. The part alone accounts for 0.012, the cofactor alone for 0.272, and on the fitted values the product $u \cdot v$ is uncorrelated with the residuals ($r = 0.001$). The values of v are reproducible up to a reliability of 0.864, which bounds what any explanation of v can reach; on the 3061 cofactors with at least 4 gap values, the size and the prime factors of c account for 0.425 of v on held-back cofactors in one split, against a ceiling of 0.898 measured on the same cofactors.

The lemma therefore explains why the most frequent values can be powerful, not how often a value occurs: in this model the powerful part alone accounts for 0.012 of the variance and the squarefree rest for 0.272. This is a measurement on a finite range and not a statement about all gaps. A version that splits g over all its powerful divisors instead of the canonical one fits the counts of the observed pairs (g, gcd) closely, but does not predict new gap values significantly better than the plain arithmetic features (0.319 ± 0.012 against 0.307 ± 0.004), because about half of its cofactors occur at a single gap; the record of this negative result is part of the deposit.

In words. For the gap values that occur at least 20 times, knowing the powerful part and the squarefree rest of a gap predicts about half of the variation in how often it occurs, and the rest matters more than the powerful part. The prediction was tested on gaps that the fit had not seen. It describes the data up to 10^{14} ; it is not a law.

Remark III.1.5 (common factors of pairs). [measured] In a sample of every 54th consecutive pair up to 10^{14} , 401 176 pairs in total, the two members are coprime in 47.69% of the cases. The most frequent common divisors are 4, 9, 8, 16, 25, 36, 27, 32, all powerful, as Lemma III.1.1 requires.

In words. About half of the neighboring pairs of powerful numbers share a factor, and the shared factor is always powerful.

We also computed the number variance of the powerful numbers in windows of up to 50 mean spacings. It lies within about 1.2% of a sum of independent lattice variances, and the large-window form of that sum grows like $L^{2/3}$ with the constant of Chan's theorem [2, Theorem 2]; in our windows the measured growth, a slope of 0.63 on a logarithmic scale, has not yet reached the exponent $2/3$. This is a check of Chan's theorem and not a new result.

Prior art and its reach. Apart from the lemma, we found no work on the distribution of the gap values between powerful numbers. We checked the work of Bajpai, Bennett and Chan in full, the OEIS entries A076446 and A001694, and records on Zenodo; the citing works of Bajpai, Bennett and Chan were queried in two databases, which returned three and none. The reach of this search is set by these databases.

2 Powerful values of cyclotomic polynomials

Remark I.4.13 in [Part I] explains where these values come from. If a member of a triple is a proper power x^r with $r \geq 2$, its neighbors $x^r - 1$ and $x^r + 1$ are products of values $\Phi_e(x)$ and $\Phi_e(-x)$ of cyclotomic polynomials. Two such values share only primes that divide $2r$, so

a value that is coprime to the others must itself be powerful. This section records what we know about powerful values $\Phi_d(x)$ at integers $x \geq 2$. For $d = 3, 4$ and 6 the polynomial is quadratic, and the question belongs to the study of powerful values of quadratic polynomials by De Koninck, Doyon and Luca [4].

Remark III.2.1 (the values $x^2 + 1$). [computed] A powerful number that is not a square can be written as a^2b^3 with $b > 1$ squarefree, so $x^2 + 1 = \Phi_4(x)$ is powerful exactly when $x^2 - b^3y^2 = -1$ has a solution for some squarefree $b > 1$; this is the form (1)–(2) of De Koninck, Doyon and Luca [4, p. 2], and their Table 1 gives the smallest case, $682^2 + 1 = 5^3 \cdot 61^2$ [4, p. 8]. For each such b the solutions form an infinite family, the odd powers of the smallest one. We solved the equation for all 182 squarefree b with $1 < b \leq 300$: it has a solution for 34 of them. The list of OEIS A175155 [10] of the values x with $x^2 + 1 = a^2b^3$ begins 682, 1 268 860 318, 1 459 639 851 109 444.

In words. The number $x^2 + 1$ is powerful for infinitely many x , but such x are very sparse: after 682 the next value in the list of OEIS A175155 is above a billion. Each of them comes from a Pell-type equation, and only some of these equations have solutions.

Proposition III.2.2 (two explicit chains of powerful values of Φ_3 and Φ_6). [proved] *Let the map $(y, m) \mapsto (127y + 672m, 24y + 127m)$ act on the starting points $(37, 7)$ (the first chain) and $(5, 1)$ (the second chain), and let (y_j, m_j) be the point after j steps. In both chains $y_j^2 - 28m_j^2 = -3$ and y_j is odd for every j ; $7 \mid m_j$ exactly when $j \equiv 0 \pmod{7}$ in the first chain and exactly when $j \equiv 6 \pmod{7}$ in the second; and the two chains contain every solution of $y^2 - 28m^2 = -3$ with $y, m > 0$. For every j with $7 \mid m_j$ the number $x = (y_j - 1)/2$ satisfies $\Phi_3(x) = 343(m_j/7)^2$, which is powerful, and $\Phi_6(x + 1) = \Phi_3(x)$. In particular $\Phi_3(x)$ and $\Phi_6(x)$ are powerful for infinitely many x .*

Proof. Since $127^2 - 28 \cdot 24^2 = 1$, the map preserves $y^2 - 28m^2$, and $37^2 - 28 \cdot 7^2 = 5^2 - 28 \cdot 1^2 = -3$. It preserves the parity of y , because 672 is even and 127 is odd. Modulo 7 it fixes y , because $672 \equiv 0$ and $127 \equiv 1$, and it adds $24y \equiv 3y$ to m . In the first chain $y_0 \equiv 2$ and $m_0 \equiv 0 \pmod{7}$, so $m_j \equiv 6j \pmod{7}$, which vanishes exactly when $j \equiv 0 \pmod{7}$; in the second $y_0 \equiv 5$ and $m_0 = 1$, so $m_j \equiv 1 + 15j \equiv 1 + j \pmod{7}$, which vanishes exactly when $j \equiv 6 \pmod{7}$. For the completeness, let $\alpha = y + z\sqrt{7}$ with $y, z > 0$ and $y^2 - 7z^2 = -3$. Its conjugate is $-3/\alpha$, so $2y = \alpha - 3/\alpha > 0$ gives $\alpha > \sqrt{3}$. Dividing by $\varepsilon = 8 + 3\sqrt{7}$ keeps the norm, and keeps the quotient above $\sqrt{3}$ as long as $\alpha > \sqrt{3}\varepsilon$; so $\alpha = \beta\varepsilon^n$ with $n \geq 0$ and $\sqrt{3} < \beta \leq \sqrt{3}\varepsilon$. For $\beta = y' + z'\sqrt{7}$ this gives $z' = (\beta + 3/\beta)/(2\sqrt{7}) < \sqrt{3}(\varepsilon + 1)/(2\sqrt{7}) < 6$, and among $1 \leq z' \leq 5$ only $z' = 1$ and $z' = 2$ make $7z'^2 - 3$ a square, so $\beta = 2 + \sqrt{7}$ or $\beta = 5 + 2\sqrt{7}$. Multiplying by ε replaces (y, z) by $(8y + 21z, 3y + 8z)$, which swaps the parities of y and z ; so $z = 2m$ is even exactly for $(2 + \sqrt{7})\varepsilon^n$ with n odd and for $(5 + 2\sqrt{7})\varepsilon^n$ with n even. Since $(2 + \sqrt{7})\varepsilon = 37 + 14\sqrt{7}$ and $\varepsilon^2 = 127 + 48\sqrt{7}$, these are the two chains, and multiplying by ε^2 is the map. Finally $4\Phi_3(x) = (2x + 1)^2 + 3 = y_j^2 + 3 = 28m_j^2$, so $\Phi_3(x) = 7m_j^2$, and $\Phi_6(x + 1) = (x + 1)^2 - (x + 1) + 1 = \Phi_3(x)$. $\square$

In the first chain the first member is $x = 18$ with $\Phi_3(18) = 7^3$; the next three have 19, 35 and 52 digits. The value $x = 18$ is the entry $k = 3$, $n = 37$ of Table 1 of De Koninck, Doyon and Luca, since $37^2 + 3 = 4\Phi_3(18)$ [4, p. 8], and they observe that for a quadratic polynomial one powerful value of this kind gives infinitely many [4, p. 7]. Their reduction on that page, applied to Φ_3 with squarefree part 7, leads in our normalization to the equation

$y^2 - 28m^2 = -3$ with $7 \mid m$. We did not find there the explicit recurrence, the two chains and their completeness, the parity of y_j , the period 7, or the transfer to Φ_6 . A direct search over all $m \leq 10^6$ finds no solution outside the two chains, as the proposition says. The two chains are disjoint: going forward from $(5, 1)$ the values of y are $5, 1307, \dots$, going backward they are negative from $(-37, 7)$ on, so $(37, 7)$ is not on the second chain. The prime x with $\Phi_3(x) = 7^3 a^2$ reported by Lavi [7], which we know from the description of his record, is the member $j = 13$ of the second chain. Other squarefree parts give other powerful values: the second value for $d = 3$ in Table 3 comes from the equation $y^2 - 52m^2 = -3$ with $13 \mid m$, that is, with 13 in place of 7.

In words. The polynomials $x^2 + x + 1$ and $x^2 - x + 1$ take powerful values infinitely often. A Pell equation produces them one after another, in two chains, and every seventh step of each chain gives one. The idea that one solution gives infinitely many is in the literature; we did not find the explicit chains there.

Proposition III.2.3 (powerful values up to a bound). [computed] *Among the 3 004 199 pairs (d, x) with $3 \leq d \leq 12$, $x \geq 2$ and $\Phi_d(x) \leq 10^{12}$, exactly 6 give a powerful value $\Phi_d(x)$. They are listed in Table 3.*

d	x	$\Phi_d(x)$	factorization
3	18	343	7^3
3	88 916	7 906 143 973	$7^2 \cdot 13^3 \cdot 271^2$
4	682	465 125	$5^3 \cdot 61^2$
5	3	121	11^2
6	19	343	7^3
6	88 917	7 906 143 973	$7^2 \cdot 13^3 \cdot 271^2$

Table 3: The powerful values $\Phi_d(x)$ with $3 \leq d \leq 12$ and $\Phi_d(x) \leq 10^{12}$ (Proposition III.2.3). The pairs for $d = 3$ and $d = 6$ are related by $\Phi_6(x + 1) = \Phi_3(x)$.

In words. Up to the bound, powerful cyclotomic values occur only for $d = 3, 4, 5$ and 6, and only 6 times. The case $d = 5$ is $3^4 + 3^3 + 3^2 + 3 + 1 = 121 = 11^2$, a square.

Remark III.2.4 (a class without squares). [computed] The numbers $x_n = (x^n - 1)/(x - 1)$ satisfy $x_{n+2} = (x + 1)x_{n+1} - x x_n$, a recurrence with $b = x + 1$ and $c = -x$ coprime and $b^2 + 4c = (x - 1)^2 > 0$. If $x \equiv 2 \pmod{4}$, then $c \equiv 2 \pmod{4}$, and Granville's theorem [5, Theorem 3] gives, for every $d \notin \{1, 2, 3, 6\}$, a primitive prime factor of x_d that divides it to an odd power. Such a prime divides $\Phi_d(x)$ and no $\Phi_e(x)$ with $e \mid d$, $e < d$, so $\Phi_d(x)$ is not a square. It can still be powerful, since an odd exponent can be 3. We checked the consequence: for $x \equiv 2 \pmod{4}$, $d \in \{4, 5, 7, \dots, 12\}$ and $\Phi_d(x) \leq 10^{14}$ no value is a square, while outside the class the square $\Phi_5(3) = 11^2$ appears at once, and for these degrees it is the only square outside the class up to 10^{14} . Inside the class, $\Phi_4(682) = 5^3 \cdot 61^2$ is powerful without being a square.

In words. For arguments of the form $4j + 2$, a theorem of Granville shows that the cyclotomic values are never squares, apart from four small degrees. Our check up to 10^{14} agrees with it, and the example $\Phi_5(3) = 121$ shows that the condition on the argument cannot simply be dropped. The theorem says nothing about cubes, so it does not decide whether such a value can be powerful.

Proposition III.2.5 (the simplest case of Φ_8). [proved] *There are no positive integers x and a and no prime $q < 2 \cdot 10^{11}$ with $x^4 + 1 = q^3 a^2$.*

Proof. For $q = 2$ there is nothing to prove, since $x^4 + 1 \equiv 1$ or $2 \pmod{16}$ while $8a^2 \equiv 0$ or $8 \pmod{16}$. Let q be odd and write $x^4 + 1 = qy^2$ with $y = qa$. By Cohn [3, p. 1347 and Theorem 1], a solution requires $q \equiv 1 \pmod{8}$ and a solution of $v^2 - qu^2 = -1$; if $\alpha = c + b\sqrt{q}$ is the smallest one, $u_n = (\alpha^n - \bar{\alpha}^n)/(2\sqrt{q})$ and $v_n = (\alpha^n + \bar{\alpha}^n)/2$, then $x^2 = v_A$ and $y = u_A$, where A is the squarefree part of c . Expanding α^n gives $u_n \equiv n c^{n-1} b \pmod{q}$. Since $c^2 \equiv -1 \pmod{q}$, the prime q divides neither c nor A , so $q \mid y = u_A$ forces $q \mid b$. For $q \equiv 1 \pmod{8}$ the fundamental unit $(t + w\sqrt{q})/2$ of $\mathbb{Q}(\sqrt{q})$ has t and w even, because $t^2 - qw^2 = \pm 4$ is impossible with t and w odd; so it lies in $\mathbb{Z}[\sqrt{q}]$, equals α , and $b = w/2$. Hence a solution requires $q \mid w$. The conjecture of Ankeny, Artin and Chowla says that this never happens, and it has been verified for all primes $q \equiv 1 \pmod{4}$ below 10^{11} [13] and between 10^{11} and $2 \cdot 10^{11}$ [14]. $\square$

We checked the last step independently for the 2384 primes $q \equiv 1 \pmod{8}$ below 10^5 : in none of them does q divide b . The proposition covers only the case in which the squarefree part of $x^4 + 1$ is a single prime. For small squarefree parts the literature settles the question completely: Cohn solved $x^4 + 1 = Dy^2$ for all squarefree $D < 10^5$, and apart from $D = x^4 + 1$ with $y = 1$ the only solution is $261^4 + 1 = 70\,258 \cdot 257^2$ [3, abstract]. If $x^4 + 1$ is powerful and D is its squarefree part, then $D \mid y$; neither case allows this, so the squarefree part of a powerful $x^4 + 1$ is at least 10^5 .

In words. The number $x^4 + 1$ is the eighth cyclotomic value. If it were a cube of a prime times a square, a theorem of Cohn would force a very special divisibility in the smallest solution of a Pell equation, and that divisibility is exactly what a well-tested conjecture forbids; computer searches by others confirm it far enough to exclude all primes below $2 \cdot 10^{11}$.

3 An independence test along the divisors of the index

The heuristic of Part I treats the event that a prime of rank d divides $T_d(m)$ at least twice as a rare event of probability about $1/p$, independent of other such events (Proposition I.6.1 in [Part I]). Part I tests independence across the primes of one kernel. A lattice point (m, k) with many divisors carries one such event for each divisor d of k , and the divisors are linked: qd is a multiple of d . This section tests whether the events at d and at qd are coupled.

Remark III.3.1 (no joint event along the divisors, and the power of the test). [measured] For the 25 kernels with the most lattice points in the box of Theorem I.3.1 in [Part I], we took each rank d and its smallest prime p of rank d below $2 \cdot 10^6$, and set the status of d to 1 if $p^2 \mid T_d(m)$ and to 0 otherwise. We then formed all pairs (d, qd) with $q \in \{3, 5, 7, 11, 13\}$ for which both ranks have such a prime: 294 920 pairs. The status is 1 at d alone in 38 pairs, at qd alone in 8, and at both in 0; independence predicts 0.001 pairs with both. To see what this test could detect, we simulated a coupling in which the status at qd copies the status at d with probability ρ , independently for each pair: for $\rho = 0.1$ one expects about 3.8 copied pairs, and in at least 98% of 400 simulations at least one joint event appeared, against less than 1% for $\rho = 0$.

In words. If the rare event at one rank made the same event at a multiple rank more likely, pairs with the event at both ranks would appear. There are none. The test is not blind: a coupling in which one time in ten the second event copies the first would almost always have shown up, and one of half that strength would still have shown up in most runs. A much weaker coupling, or one that makes the second event less likely, could not be seen with so few events, and the test uses only the smallest prime of each rank.

The test is formally a χ^2 test on the table, but with fewer than one expected joint event the statistic only records whether a joint event occurs; the simulation measures exactly that. One event at a rank d enters a pair for each q with qd occupied, so the 38 pairs come from fewer distinct events; the simulation copies pair by pair and therefore overstates the power against a coupling that acts once per event. In the prior-art search for this part (arXiv, Zenodo, Google Scholar and OEIS, at the level of titles and abstracts) we found no earlier test of independence along the divisors of the index.

4 Wieferich fractions of Pell units

For a squarefree m and a prime $p \nmid 2m$ let $e = \left(\frac{m}{p}\right)$ and write $\varepsilon_m^{p-e} = A + B\sqrt{m}$ modulo p^2 . Then $A \equiv 1$ and $B \equiv 0 \pmod{p}$, and we call $y_p = ((B/p) \bmod p)/p \in [0, 1)$ the *Wieferich fraction* of ε_m at p . It vanishes exactly when the order of ε_m modulo p^2 equals its order modulo p ; for a prime with a rank this is the condition $p^2 \mid T_{\alpha_m(p)}(m)$ of Part I. In Katz's language [6], ε_m is a point on the torus $x^2 - my^2 = 1$, and y_p is its Wieferich fraction for the framing whose Lie lattice is $\mathbb{Z}\sqrt{m}$, which Katz describes [6, Section 6]. His Framed Wieferich Conjecture [6, Conjecture 6.1] predicts that such fractions are equidistributed.

Remark III.4.1 (no deviation from equidistribution, memory or coupling). [measured] For the 25 kernels of Section 3 and all primes $p < 3 \cdot 10^6$ with $p \nmid 2mU_1$, that is, for 5 420 294 pairs (m, p) , we computed y_p .

- (a) *Near zero.* For six thresholds t from 10^{-2} to 10^{-7} the number of $y_p < t$ agrees with the exact expectation for uniform y_p ; the largest deviation is 1.37 standard deviations. For example, $y_p < 10^{-4}$ holds 593 times against 583.9 expected, and $y_p = 0$ holds 56 times against $\sum 1/p = 50.71$. Only 16 of these zeros lie at primes that divide some $T_k(m)$, which happens exactly when the order of ε_m modulo p is divisible by 4; they are the Wieferich events of Part I on these kernels (Remark I.6.11 in [Part I]), and the other zeros do not appear there. Split by the 2-adic valuation of the order of ε_m modulo p (odd, 2 mod 4, 0 mod 4) and by e into 6 classes, the smallest p -value of a χ^2 test of uniformity is 0.014, or 0.08 after the Bonferroni correction.
- (b) *Along the primes.* For each kernel, the correlation of y_p with the fraction j primes later, for $j = 1, \dots, 20$, gives 500 tests; 7 exceed $|z| = 2.576$ against about 5 expected, the largest is 3.30, with Bonferroni $p = 0.48$. A memory planted in test data, with 5 percent of the values replaced by their predecessor, is found with $z = 21.4$.
- (c) *Between kernels.* For the 300 pairs of kernels, the correlation of their fractions at common primes has largest $|z| = 3.61$, with Bonferroni $p = 0.09$; a planted correlation of strength 0.02 is found with $z = 10.0$.

In words. For every unit and every prime there is a number between 0 and 1 that measures how close the prime comes to being a Wieferich prime for that unit. Over more than five million such numbers, on 25 units and the primes below $3 \cdot 10^6$, our tests find no deviation from independent uniform random numbers: near zero, from one prime to the next, and from one unit to another. Katz’s conjecture predicts (a); applied to the product of two such tori it also predicts (c); (b) goes beyond it. The only computation of Wieferich fractions that his paper mentions is one for elliptic curves, by Bombieri, reported as compatible with the conjecture. Censuses of the zeros $y_p = 0$ exist for single units, for example Ryan’s for $1 + \sqrt{2}$ up to 10^9 [11, Sections 2.1 and 5.1]; we did not find a test of the distribution of the fractions themselves, near zero, along the primes or between units. A finite test cannot prove equidistribution; it shows no deviation at the sizes the controls can detect.

5 Questions

Question III.5.1 (the most frequent gap). [open] Let $g(x)$ be the most frequent gap between consecutive powerful numbers up to x . Is $g(x)$ powerful for all large x , and does the share of powerful values among the N most frequent gaps up to x tend to 1 when N is fixed and $x \rightarrow \infty$?

Proposition III.1.2 and Table 1 are consistent with the first part at three bounds up to 10^{14} . For the second part they point both ways: the share of powerful values among the two hundred most frequent rises, while the leading block in which all values are powerful shrinks. Lemma III.1.1 gives powerful gaps an advantage; it does not show that the advantage wins in the limit.

In words. Is the most common distance between neighboring powerful numbers always itself powerful, however far one looks? Up to 10^{14} it is, but the data do not show a clear trend for the values just below the top.

Question III.5.2 (powerful values of $x^4 + 1$). [open] Is $x^4 + 1$ powerful for some integer $x \geq 1$?

By Proposition III.2.5 and the paragraph after it, the squarefree part of such a value is at least 10^5 , and it is not a prime below $2 \cdot 10^{11}$. For $x^2 + 1$ the answer is yes (Remark III.2.1).

In words. Can the eighth cyclotomic value $x^4 + 1$ ever be powerful? The fourth one, $x^2 + 1$, can. For $x^4 + 1$ only special shapes are excluded so far.

Remark III.5.3 (a conjecture on the most frequent gap). [heuristic] We state a weak and a strong form for the most frequent gap $g(x)$ of Question III.5.1. *Weak form:* $g(x)$ tends to infinity with x , and $g(x)$ is powerful for all large x . *Strong form:* for all large x , $g(x)$ is the square of a product of the first primes, $(2 \cdot 3 \cdots p_r)^2$, with r growing with x . The strong form implies the weak one; the weak one claims only what the data and Lemma III.1.1 support directly, and the strong one also names the shape. Up to 10^{12} the most frequent value is $900 = (2 \cdot 3 \cdot 5)^2$, and up to 10^{13} and 10^{14} it is $44100 = (2 \cdot 3 \cdot 5 \cdot 7)^2$ (Table 1). The reason we expect it is Lemma III.1.1: if $g = P^2$ with P squarefree and $P^2 \mid n$, then $n = P^2 m$ and $n + g = P^2(m + 1)$, so the primes of P already occur at least twice in both numbers, and only the other primes of m and $m + 1$ must occur squared. A larger P frees more primes but leaves fewer $m \leq x/P^2$; the smallest primes free the most for the smallest cost, and as x grows the balance moves to a larger P . A balance of this kind is behind the conjecture

of Odlyzko, Rubinstein and Wolf that the most frequent gap between consecutive primes eventually runs through the products of the first primes, 30, 210, 2310 [8]; we found no such statement for powerful numbers. Two values do not make a trend. The weak form fails if $g(x)$ stays bounded, or if a value that is not powerful is the most frequent gap for arbitrarily large x . The strong form fails in addition if a powerful value of another shape, such as 3600 or 3528, which follow 900 and 44 100 closely in the counts, is the most frequent gap for arbitrarily large x . Which of the two forms is right is open.

In words. We expect the most common distance between neighboring powerful numbers to keep growing and to stay powerful (weak form), and we suspect that it is always the square of $2 \cdot 3 \cdot 5 \cdots$, as 900 and 44 100 are (strong form). For the primes a similar pattern is conjectured, with 30, 210, 2310. This is an expectation supported by two values and a mechanism, not a result.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent reading of the source.

statement	title	type
Lemma III.1.1	Bajpai, Bennett and Chan [1, proof of Theorem 1.1]	literature
Proposition III.1.2	the head of the gap spectrum	computed
Remark III.1.3	a ceiling for common divisors	computed
Remark III.1.4	a counting model, measured	measured
Remark III.1.5	common factors of pairs	measured
Remark III.2.1	the values $x^2 + 1$	computed
Proposition III.2.2	two explicit chains of powerful values of Φ_3 and Φ_6	proved
Proposition III.2.3	powerful values up to a bound	computed
Remark III.2.4	a class without squares	computed
Proposition III.2.5	the simplest case of Φ_8	proved
Remark III.3.1	no joint event along the divisors, and the power of the test	measured
Remark III.4.1	no deviation from equidistribution, memory or coupling	measured
Question III.5.1	the most frequent gap	open
Question III.5.2	powerful values of $x^4 + 1$	open
Remark III.5.3	a conjecture on the most frequent gap	heuristic

The types: *proved*, proved in this part; the proof may use published results (2); *literature*, due to the cited source, sometimes with a proof given here for completeness (1); *computed*, the result of an exact computation, or a proof that uses one of our computations (5); *measured*, a measurement, with a model that is not proved (4); *heuristic*, a heuristic, used as a yardstick only (1); *open*, an open question (2).

Data and code availability

Every number of this part that we computed is generated from a result file of a data deposit or from a cited constant; no computed number is typed by hand. The data deposit is archived with the collected volume, DOI 10.5281/zenodo.23127547.

What the deposit contains. The deposit has two parts. The core holds what the text cites: (i) the result files, in JSON and plain text, behind every computed number, together with the raw output of the scripts that wrote them; (ii) the scripts, written in Python, each stating before it runs what it expects to find and the positive and negative controls that it runs; (iii) a manifest with the SHA-256 hash and the size of every file, which the build checks before it reads a result file; (iv) the decimal expansions of the building blocks of Part I that the searches left open (Proposition I.6.8 in [Part I]), including those that the algebraic split of Remark I.6.13 in [Part I] settles; (v) the certificates and the second routes for the computations that settle a building block; and (vi) the file `OPEN_PROBLEMS.md`, which lists the open problems of the volume. The archive holds the remaining scripts and results of the investigation, including negative results that the text does not print.

Where the files are. The collected volume and its three parts are published as four linked records on Zenodo, and the same files are in one repository on GitHub. The record of the collected volume (DOI 10.5281/zenodo.23127547) holds the volume and the whole deposit, the core and the archive described above. The record of each part holds that part and the result files that its text cites, and links to the record of the volume. The repository (github.com/berkay-yueksel-sayim/erdos-364-365-powerful-numbers-pell) holds the four PDFs and the deposit, and its README lists the four records. The records of Parts I, II and III have the DOIs 10.5281/zenodo.23127703, 10.5281/zenodo.23127705 and 10.5281/zenodo.23127707.

How the numbers get into the text. The source of the text refers to each computed number by a key. A build tool reads the result file named for the key, verifies it against the manifest, computes the number in its printed form, and writes it into the text; it stops if a key has no source or if a file differs from the manifest. The same tool generates the table of the types of the statements and the list of open problems from the source of the text, and it is part of the core.

Software. The scripts use Python 3.12 with gmpy2 2.3, SymPy 1.14 and NumPy 2.4. The elliptic curve method and its relatives are run with GMP-ECM 7.0.6 [15], used as an installed program inside a container image built from the Debian package and not part of the deposit; the recipe of the image is in the core. The elliptic curve certificates for primality were produced in the same way with PARI/GP 2.17.2 [12] and are part of the deposit; a separate program of ours checks each of them, so the result does not depend on that program. Further work is planned to support these certificates with a second prover written independently by us. The text is set with pdf $\text{\TeX}$.

Licenses. The text and the data are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), and the code under the Apache License, Version 2.0. Programs of others that the scripts use as tools, such as GMP-ECM, are not part of the deposit and keep their own licenses.

Use of AI tools

Generative AI tools were used throughout this work: large language models of the Claude family by Anthropic, mainly Claude Opus 5 and 5.5, and also Claude Fable 5.1 and Claude Sonnet 5.5 and 5, between August and October 2026, through the Claude Code environment. They were used to draft the text and the mathematical arguments, to write the computer programs, to run and check the computations, and to search for and read the literature. The author works independently, is not trained as a number theorist, and learned the subject actively during the work; the tools made it possible to carry out and check computations on this scale. The author led the work in stages: each stage was set by a prompt that stated what it should achieve, within a stage the tools followed a plan agreed with the author, and the author considered the results before giving the next prompt, revising the plan when the results required it. The author contributed ideas where they helped and framed how a problem should be attacked, decided which statements to keep, and is responsible for the content. The programs of the project, including its own checking and build tools, were developed with the AI tools, and the author started the long computations, such as the factor searches, on the author's own machine.

Several parts of the checking do not rest on these tools. The elliptic curve primality certificates were produced by PARI/GP 2.17.2 and are verified by a separate program, and the factor searches use GMP-ECM 7.0.6. Every computed number is generated by a program from deposited data or from a cited constant, and the programs carry positive and negative controls. Proofs, numbers and citations were also checked in independent passes by further instances of the AI tools that were given no access to the drafting context. These tools are not authors.

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